

# Yihan Li

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## Education

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| <b>Sun Yat-sen University</b>                                                 | Sep 2023 – Jul 2027 (Expected) |
| <i>B.A. in English Language &amp; Literature, School of Foreign Languages</i> | <i>Major GPA: 3.8/4.0</i>      |
| <i>Minor in Intelligent Unmanned Systems</i>                                  | <i>GPA: 4.0/4.0</i>            |
| <i>Minor in Computer Science &amp; Technology</i>                             | <i>GPA: 3.8/4.0</i>            |

## Publications

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| <b>Anticipating Contact for Visuo-Tactile Diffusion Policies</b>                                                              | under review |
| Yating Feng, Yuhang Zheng, Lixin Xu, <b>Yihan Li</b> , Jiaju Yin, Renjing Xu                                                  |              |
| <b>Goldsmith: Gold-Loss-Guided Prompt Training for Agentic Annotation</b>                                                     | under review |
| <b>Yihan Li</b> , Hanyi Zhang, Xiaoxi Jiang, Man Guo                                                                          |              |
| <b>Unordered Landmark Visual Navigation</b>                                                                                   | under review |
| Hao Ren, Junzhe Zhu, <b>Yihan Li</b> , Zetong Bi, Le Zheng, Zhi Li, Yiqing Yuan, Zhaoliang Wan, Lu Qi, Hui Cheng              |              |
| <b>OmniTrav: A Dataset and Benchmark for 360° Vision-based Traversability via Foundation Models</b>                           | under review |
| <b>Yihan Li</b> , Hao Ren, Zhenglan Jiang, Junzhe Zhu, Hui Cheng                                                              |              |
| <b>RAPID Hand: A Robust, Affordable, Perception-Integrated, Dexterous Manipulation Platform for Generalist Robot Autonomy</b> | NeurIPS 2025 |
| Zhaoliang Wan, Zetong Bi, Zida Zhou, Hao Ren, Yiming Zeng, <b>Yihan Li</b> , Lu Qi, Xu Yang, Ming-Hsuan Yang, Hui Cheng       |              |

## Work Experience

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| <b>HCL Lab (HKUST(GZ))</b>                                                                                                                                                                                                                                                                                                                     | Jan 2026 – Present      |
| <i>Research Assistant</i>                                                                                                                                                                                                                                                                                                                      | <i>Prof. Renjing Xu</i> |
| <ul style="list-style-type: none"><li>Conducting dexterous-hand reinforcement learning research, with emphasis on Real-to-Sim transfer pipelines for reproducible policy training and evaluation.</li><li>Developing and validating VLA-style multimodal manipulation methods for visuo-tactile dexterous tasks.</li></ul>                     |                         |
| <b>Yun Drone</b>                                                                                                                                                                                                                                                                                                                               | Jun 2025 – Mar 2026     |
| <i>UAV Development Intern</i>                                                                                                                                                                                                                                                                                                                  |                         |
| <ul style="list-style-type: none"><li>Selected UAV propulsion components and participated in formation-flight development, navigation and localization, and flight-test workflow design.</li><li>Explored RL-based disturbance-resilient control and analyzed flight logs to support anti-disturbance tuning and system iteration.</li></ul>   |                         |
| <b>RAPID Lab (SYSU)</b>                                                                                                                                                                                                                                                                                                                        | Nov 2024 – Jun 2025     |
| <i>Research Assistant</i>                                                                                                                                                                                                                                                                                                                      | <i>Prof. Hui Cheng</i>  |
| <ul style="list-style-type: none"><li>Developed the dexterous-hand control stack, visuo-tactile temporal alignment, and MuJoCo Sim2Real validation workflow for manipulation experiments.</li><li>Built visual obstacle-avoidance and topological-map planning models, connecting perception outputs with embodied navigation tasks.</li></ul> |                         |
| <b>RoboTech Association (SYSU)</b>                                                                                                                                                                                                                                                                                                             | Sep 2024 – Present      |
| <i>President; Competition Team Captain</i>                                                                                                                                                                                                                                                                                                     |                         |
| <ul style="list-style-type: none"><li>Led 160+ members and competition teams, coordinating training schedules, technical reviews, and cross-group collaboration.</li><li>Authored robotics onboarding tutorials across mechanical design, hardware, embedded control, and algorithms.</li></ul>                                                |                         |

## Projects

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**RoboMaster Visual System Development (RMUL)** Mar 2025 – Jul 2025  
*Vice Captain; Team Advisor; Algorithm Group Leader for Navigation and Vision*

- Built Jetson/NUC vision with EKF auto-aiming (90%+ hit rate) and TensorRT/OpenVINO YOLO inference at 10 ms.
- Led navigation and vision algorithm planning, STM32 communication, and closed-loop deployment for competitive robotics.

**RAPID Hand: Perception-Integrated Dexterous Manipulation Platform** Nov 2024 – Jun 2025  
*Research Assistant*

- Implemented drivers, visuo-tactile fusion, MuJoCo Sim2Real training, and Diffusion Policy tasks with 50/50 success.
- Integrated 1 kHz hand control, wrist vision, fingertip tactile sensing, and joint proprioception under 7 ms synchronization latency.

**OmniTrav: 360° Vision-based Traversability Benchmark** Jun 2025 – Mar 2026  
*First Author*

- Proposed C2T RGB traversability and built a 4.7k-scene, 250k-frame, 577 GB automated foundation-model dataset.
- Evaluated pinhole, fisheye, and panoramic cameras; achieved panoramic obstacle detection F1 of 1.0 and pinhole median error of 0.27 m.

**GenreShift and Rosetta: LLM Systems for Linguistic Research** Oct 2024 – Nov 2025  
*Project Leader*

- Built GenreShift (corpus, LoRA, MoE, patent No. 2025105553019) and Rosetta (Streamlit/Docker annotation with Kimi/DeepSeek APIs).
- Supported interactive concept management, data import/export, and LLM-assisted academic discourse annotation.

## Awards

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**National First Prize (Team Captain)** 2025  
China Robot and Artificial Intelligence Competition

**Overall National Champion (Team Captain)** 2025  
National Intelligent Unmanned Systems Application Challenge

**National Second Prize (Team Captain)** 2025  
National Undergraduate Electronics Design Contest

**National Second Prize (Vision Group Leader)** 2025  
RoboMaster University League

## Skills

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|-----------------------------------|---------------------------------------------------------------------------------------------------------------|
| <b>Programming:</b>               | ROS1/2, Python, C/C++, MATLAB, STM32/Arduino.                                                                 |
| <b>Robotics/ML:</b>               | Visual navigation, 3D perception, dexterous manipulation, MuJoCo, TensorRT, OpenVINO, LLM annotation systems. |
| <b>Engineering and Languages:</b> | Fusion 360, SolidWorks, EDA design; AOPA Multi-rotor VLOS Pilot; Mandarin (native), English (TEM-4), French.  |